UNIVERSITY OF NAIROBI

INFLUENCE OF WEATHER PARAMETERS ON REPORTED CORONA VIRUS PREVALENCES IN KENYA

BY

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I10/81549/2017

A Project Report submitted in Partial Fulfillment of the Requirements for Award of the Bachelor of Science Degree in Meteorology of the University of Nairobi

June, 2021

DECLARATION

I hereby make a declaration that the work contained here is my original work. It has not been copy pasted from any other source or student except due to reference or acknowledgment is made explicitly in the text nor has any part been written for me by another individual. The work has neither been presented to this university or any other academic institution.

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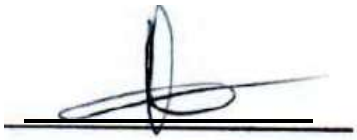
Signature:  **Date: 21/06/2021**

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DEDICATIONS

I dedicate this work to God Almighty who is my Father, my pillar, my source of inspiration, wisdom, strength, knowledge and understanding throughout this project and on his wings only have I soared. I also dedicate this work to my mum Eunice Anyango who has been encouraging me all the way and whose encouragements have made sure that I give it all it takes to finish that all I had started. To my brothers Lawrence and Antony, my sisters Medrine and Ann I thank you too.

I also dedicate this work to the department of meteorology that ensured that ensured that we got all the data needed for the project work and that ensured that we got the appropriate knowledge during our four years course. Thank you all, my love for you all can never be quantified. May God bless you.

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I again take this opportunity to thank my supervisor Dr Mutai for being open for consultation. His corrections and advises did a lot in order for this work to be realized.

ABSTRACT

The novel corona virus (SARS-CoV-2) has spread globally and nationally in the country of Kenya and was been declared a pandemic by ministry of health. While influenza virus shows seasonality, it is unknown if COVID-19 has any weather-related effects. In this study, the influence of weather parameters, temperature and windspeed, on covid-19 cases was investigated. The rural and urban aspects were considered in this study. Thus far Nairobi and Mombasa Counties were chosen to represent urban aspects while Uasin-Gishu represented the rural aspects.

The study found that wind speed had some relationship with corona virus cases in Uasin-Gishu county, it was seen that the influence would be more realized after every season where seasons that had low windspeed reported a little higher number of people who tested positive of the corona virus. However, the study did not find temperature as a major course for the number of infections in Kenyan rural areas as there was weak relationship between temperature and corona virus cases in Uasin-Gishu county.

On the other hand, the study found that temperature might have influenced the number of infections in the urban areas. Higher temperatures reported lower number in the covid-19 infections while lower temperatures reported higher number in the covid-19. Consequently, seasons with lower average temperatures reported relatively higher cases in Mombasa and Nairobi than seasons with higher temperatures which reported lower corona cases.

It was also found in the study that wind speed had a weak relationship with the number of corona virus infections in the urban areas and so it might have contributed very little to the reported cases in Nairobi and Mombasa counties.

In this study it was established that indeed weather parameter might have played a role in the corona virus cases in Kenya where temperature might have affected urban areas while windspeed might have affected the rural areas.

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LIST OF ACRONYMS

2019-nCov: is the 2019 novel corona virus

AW by koppen Geiger: is the tropical savanna climate with dry winter

CDC: center for disease control

Cfb: according to koppen is a temperature oceanic climate

Covid-19: is the coronavirus disease of 2019

DJF: is the season for December, January, February.

JJA: is the season for June, July, August.

MAM: is the season for March, April, May.

RNA on aerosol: is the Ribunucleic acid for SAR cov.

SAR Cov-1: is the sever acute respiratory syndrome coronavirus 1

SAR Cov-2: is the sever acute respiratory syndrome coronavirus 2

SON: is the season for September, October, November

WHO: is the world health organization.

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CHAPTER ONE

INTRODUCTION

1.0 Background of the Study

In the beginning of the year 2020 several cases of the novel corona virus, also known as the 2019-ncov or COVID 19 started in china in the province of Hubei in the city of Wuhan

The disease quickly spread to other regions since it highly contagious and so its high transmissive nature and many global travels it was then declared a pandemic by the World Health Organization on 11th march 2020

Human corona virus has been associated with a wide spectrum of respiratory diseases in different studies. Influenza virus have been observed to follow a seasonal pattern and SARS cov-1 (a type of corona virus), was also found to lose its ability to survive at higher temperatures which could be due to the breakdown of their lipid layer at higher temperatures.

1:corona virus photo

1:CORONA VIRUS PHOTO 1

Figure 1 shows picture of the corona virus
Source: www.bcradiology.ca

Due to the trend of high reported cases in countries with extreme low temperatures the disease can therefore be associated variation in temperatures and wind speed being that it can stay in air.

Covid-19 first case was reported in Kenya on 13th March 2020, this led to the panic among Kenyans and its government. On 15th of march 2020 the third case had been reported and the

government issued a directive on a lockdown. The Kenyan schools and churches bars and public gatherings were closed down. Despite all these the disease ravaged the country from the city of Nairobi where the first case was reported to all almost all the counties of the nation

The ministry of health would issue daily update on the newly reported cases, later there was a dusk to don curfew in the country. The latest report on covid-19 cases on 1st February 2021 the ministry of health through Dr. Rashid Aman reported that the official death toll in Kenya had reached one thousand seven hundred and seventy-three, (1,773), one hundred and one thousand (101,000) total official reported cases eighty-four thousand one hundred and fourty three (84,143) recovered cases.

Corona virus was first believed to only spread when one coughs or sneezes, that is, it was said to be in the tiny droplets that comes from one who has coughed or sneezed and that the droplet droplets were heavy that could only move to a distance not more than one meter and thus social distancing of one meter was drown.

However, on 15th October 2020 the Center for Disease control and Prevention (CDC) reported that the disease can also be spread through breath and that the virus was able to stay in the air for a minimum of 3 hours after that the existence in the air was not known to take how long

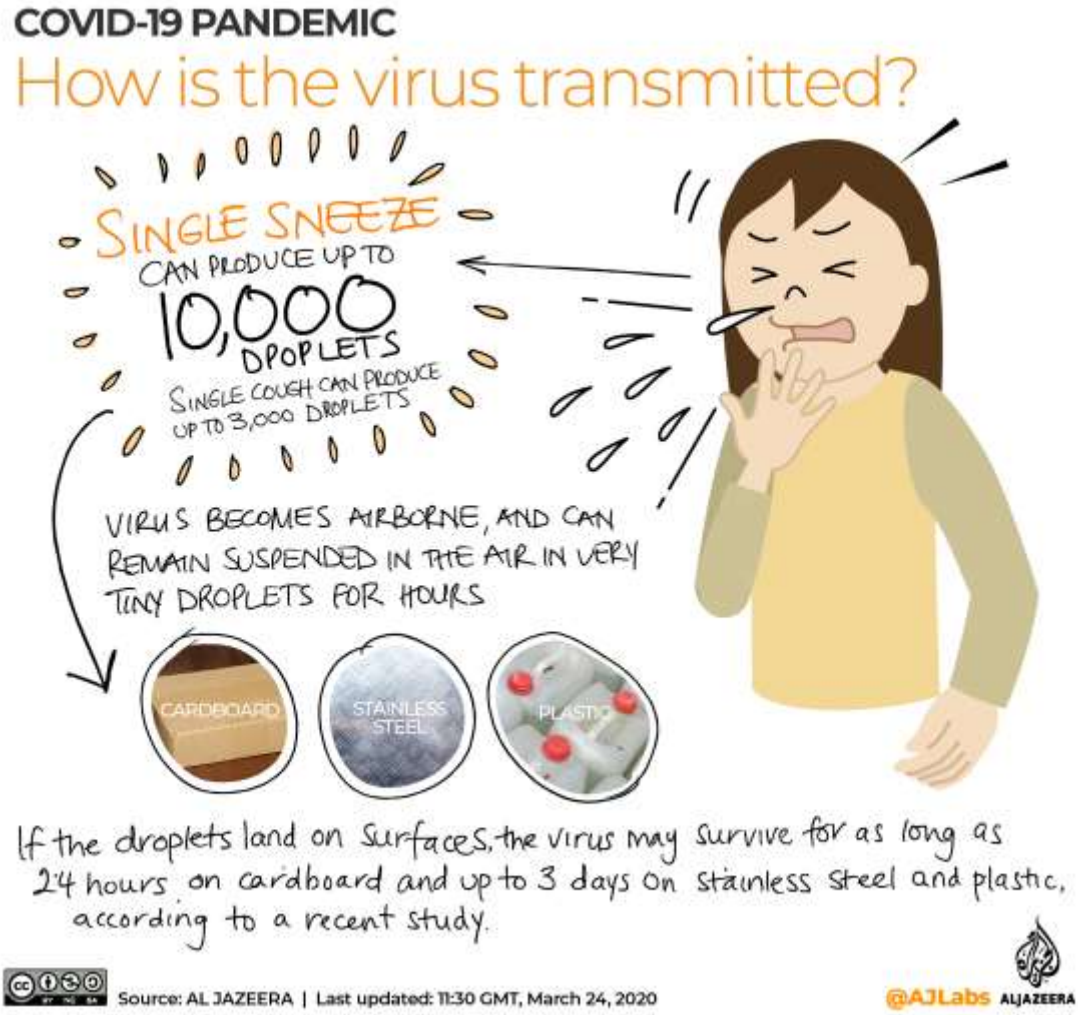

Figure 2 above shows spread of covid-19
Source: AL JAZEERA.

Nazar Raoof, M, D a division director of infection disease at Raritan bay medical center says that the best way to prevent airborne spread is through wearing of mask and practicing social distancing.

Covid-19 currently has no cure, the aim of treatment is to manage and reduce the symptoms until one is able to recover fully from the disease. Most people close to 80% have asymptomatic infections which can be treated at home. In this case one is required to self- isolate for at least one week until he or she recovers. A proportion of people who contract covid-19 will need hospital care, this likelihood is very high for people with other diseases like blood pressure, asthma, and maybe diabetes

1.1 Problem Statement

This study is to determine how temperature and wind speed has related to the spread of corona virus disease in Kenya. Reports from outside Kenya shows that approximately 85 percent of covid-19 reported cases until 1st May 2020 making approximately 3,000,000 reported cases (out of approximately 29 million tests performed) have occurred in regions with low temperature ranging between 3^0c and 17^0c

In Kenya, areas with averagely low annual temperature of about 17^0c have recorded relatively high corona virus cases, the season of JJA in Kenya reported the highest corona virus cases in the year 2020 ignoring the second wave of infection which occurred as a result of negligence of covid-19 protocols by 'mwananchi' after the cases started going down, the second wave started from October to December the year 2020.

Nairobi of annual average temperature of 17.7^0c and average wind speed of 3.583m/s recorded 43,098 cases until February 1st. On the other hand, Mombasa also a city with same settlement structure, almost same demography and almost equal factors that lead to the spread of corona virus of has an average annual temperature of 27.7^0c, average windspeed of 4.08m/s by 1st of February had total cases mounting to 9,054 confirmed cases.

It was also reported that urban areas had more covid-19 cases than rural areas, this was seen in the context that urban areas have higher demography than in rural areas which is true. However, this could also be as a result of wind speed. The flow of the wind speed in the city is characterized mean wind speed that are 20 to 30 percent lower than those of winds blowing across the rural area. This occurs because urban areas have rough terrain of tall buildings and different structures that causes friction thus lowering the wind speed than in rural areas. This makes it easier for covid-19 to spread much faster in urban areas than in rural areas. More prominently is the distinction in number of cases between countries in the tropics which recorded low case compared to countries in the extra tropics that recorded high number of covid-19 cases.

All these suggest that when corona virus is in the air, it will remain in the air endangering lives of its inhalers unless moved away by wind horizontally or vertically by increase in temperature through positive buoyancy.

Having all these proofs on the relationship between weather and the number of covid-19 infections, my study was to help reduce the covid-19 cases in enclosed places where transmission may be easy like offices, courts, classrooms, churches, conference halls and bars and also to warn the public to be scared whenever the day is chilly since the infection rate may be high.

1.2 Justification.

Since the 13th March when corona virus first case was reported in Kenya it has claimed the lives of 1,773 official deaths and infected 101,000 people all over the country, the government has put in place certain measures like quarantine of infected people, issued curfew, lockdown, wearing of mask and washing of hands including social distancing and even disinfecting places to curb the spread of the virus.

Despite all these measures that were put in place, the virus continued to ravage the country and at times created fear that the whole generation might be whipped out of the face of the earth

Although wearing of masks helped but did not solve much as some still had infections in places where there was strict adherence to covid-19 protocols among which were wearing of masks and social distancing.

The question was that, what might have led to this?

My study of how temperature and wind speed might have led to the spread of covid-19 virus will help fill the gaps which may not have been fully met by covid-19 protocols because people will be able to consider under what temperatures they are to have high risks of infection and may avoid.

1.3 Objectives

The main objective of this study was to investigate/assess the influence of weather on the spread of covid-19 in Kenya.

Specific Objectives:

1. Determine the spatio-temporal distribution of weather.
2. Determine the spatio-temporal distribution of covid-19 prevalence.
3. Establish relationship between weather and covid-19 prevalence.

1.4 Area of Study

1.4.1 Nairobi

Nairobi is the Kenya's capital situated in south-central Kenya. Nairobi lies within latitude 1.2921degrees south and longitude 36.8219 degrees east. It is 140km south of equator, it is

surrounded by several cliffs and forests. It is adjacent to the eastern edge of rift valley and to the west of the city are Ngong hills.

It has an elevation of 1.7921 km above the sea level. It has a population of 4.4 million people occupying an area of 696 square km making it to have a population density of 6317/sq km

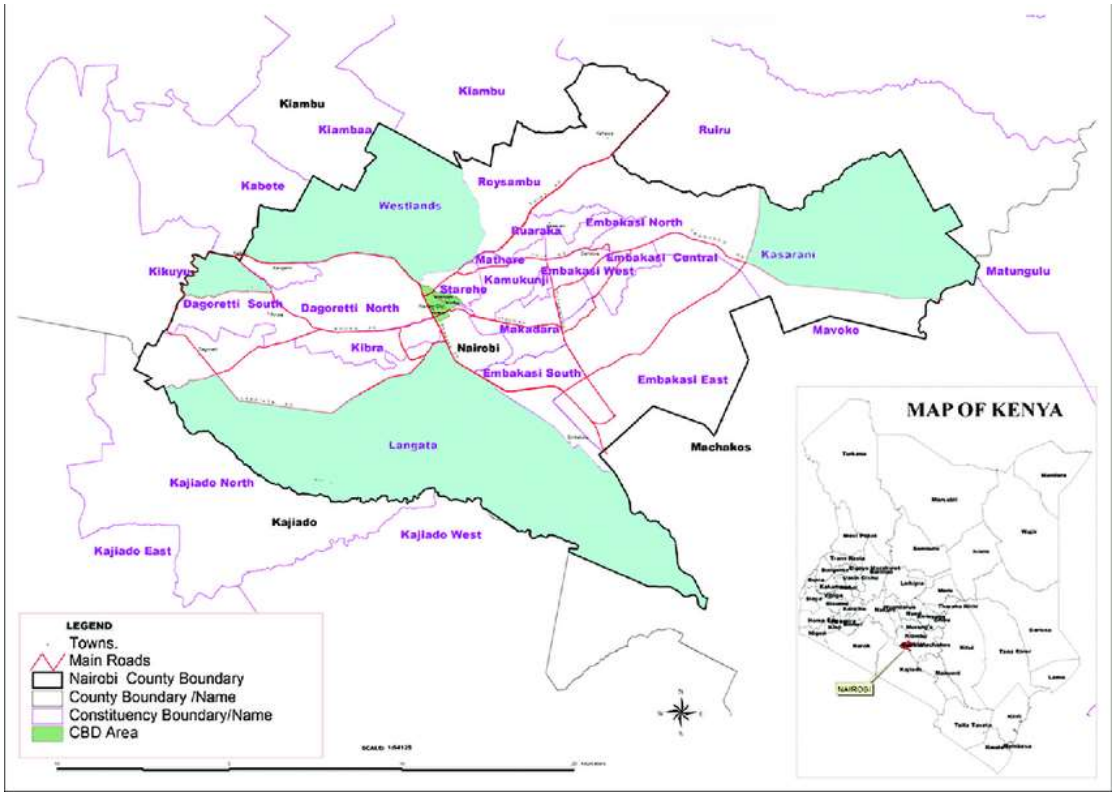

Figure 3 Map of Nairobi city
Source: www.iebc.or.ke/uploads/resources/WHXao7x83D.pdf.

1.4.2 Mombasa

Mombasa is the second largest Kenyan city found at the coastal region south east of Kenya along Indian ocean. It is the country's oldest second largest city with a population of 1.2 million according to the 2019 population census. It has an altitude of 50m above the sea level with an area of 294.7 square kilometers making it to have a population density of 4071.9/square kilometers

It is found within latitude 4.0435 degrees south and longitude 39.6682 degrees east.

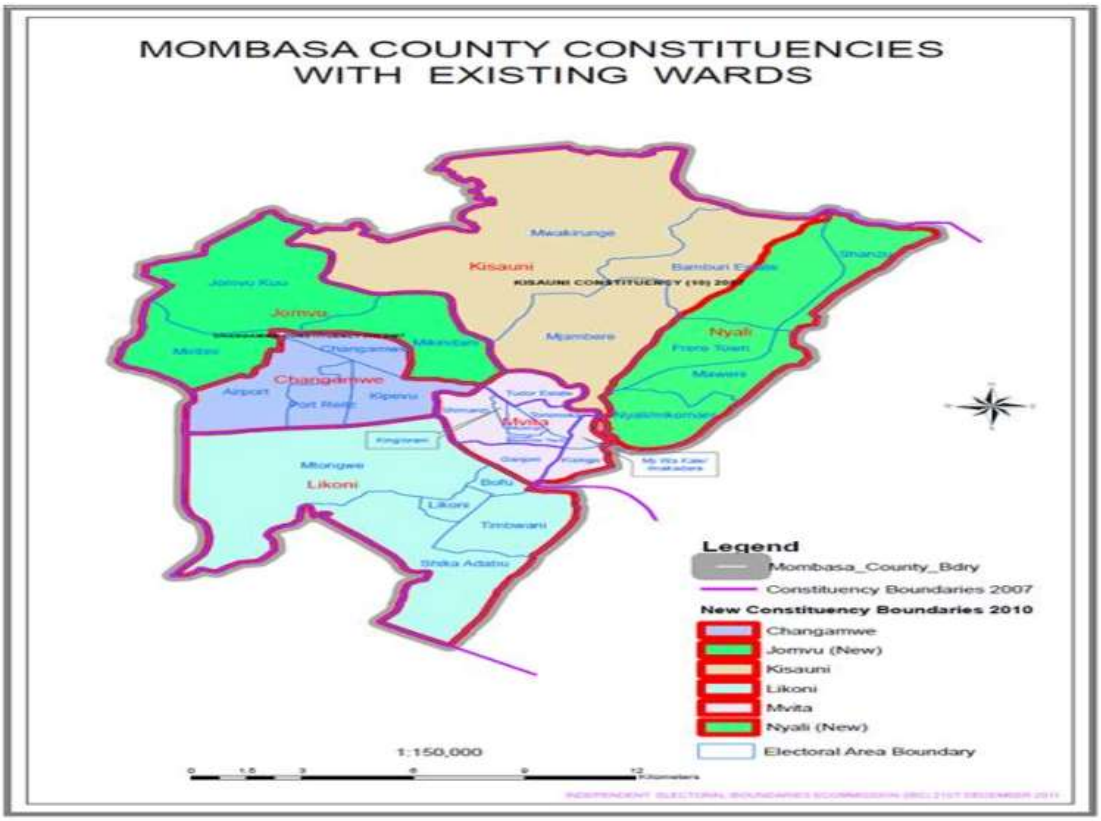

Figure 1: A Map of Mombasa County

Figure 4: A Map of Mombasa county
Source: **www.iebc.or.ke/uploads/resources/WHXao7x83D.pdf**

1.4.3 **Uasin-Gishu**

This is a county in Kenya located in the former rift-valley province. Eldoret is the main town in Uasin-Gishu is Eldoret town. I lies in latitude 0.5528 N of equator and longitude 35.3027 east. It is found 1500M elevation above the sea level. It is a highland plateau with an altitude falling from 2700M to about 1500M above the sea level.

It has a population of 1.16 million people occupying an area of 2955km squared. Density of 392.55/Square km.

Figure 5: a map of Uasin-Gishu county
Source: **www.iebc.or.ke/uploads/resources/WHXao7x83D.pdf**

1.4.4 Climate

Mombasa has a tropical climate, in winter there is less rainfall than in summer although its winter and summer are not much pronounced, the climate here is classified as AW by Koppen-Geiger system. The

temperature here is about 25.5 degrees and about 997mm of precipitation falls annually

The climate is warm and temperate in Nairobi, the climate is considered to be cfb according to Koppen-Geiger climate classification. The temperature annual average is 19.12 degrees with an annual precipitation of 674mm.

Uasin-Gishu is one of the coldest regions in Kenya with an average temperature of 17.8 degrees. July has the least sunshine hours in the county. It receives the highest rainfall in August and the least in February, has an average wind speed of 6.0m/s.

1.5.0 LIMITATIONS OF THE STUDY

1) Lack of personal data such as the details of the country, county and the travelling history of the patient tested positive of the covid-19.
2) The study did not consider 1-14 days incubation period of the corona virus.

1.5.1 ASSUMPTIONS OF THE STUDY

1) Each county of the study had their independent prevalence and that there was no inter-county transmission.
2) urban areas studied had same population density
3) Other weather parameters were kept constant and variation was only in temperature and windspeed.
4) Effect of weather parameters on covid-19 prevalence was gradual from daily to monthly until full effect after a season.

CHAPTER TWO

2.0 LITERATURE REVIEW

Zhan et al in 5th October identified airborne transmission as a dominant route for spread of covid-19, transmission of corona virus arises from coughing or sneezing of an infected person producing virus-containing droplets and aerosols, virus transmission from person through direct or indirect contact and airborne or droplet routes, large droplets mainly out of air to cause person or object contamination while aerosols are efficiently dispersed in air. Direct and airborne transmission occur in short range and extended distance or time, respectively, inhaled airborne virus deposit directly in the human respiratory truck

Recent measurements identified SAR-cov-2 RNA on aerosols at Wuhan's hospital and outdoor in northern Italy revealing the likelihood of indoor and outdoor airborne transmission.

Within enclosed environment, virus-containing aerosols from human atomization are readily accumulated, and elevate levels of airborne viruses facilitate transmission from person to person. Transmission in airborne in open air is subject loos of concentration, although virus accumulation still occurs due to the stagnation under polluted urban condition and may be still air.

They also examined ambient conditions relevant to the outbreak in Wuhan, Italy, an NYC. The initial outbreaks of covid-19 in Wuhan coincided with winter haze season in China. Winter haze conditions in China likely to exacerbated outdoor virus spreading because of low UV radiation, air stagnation (lacking ventilation on city scale), and low temperatures. Also, there may exist a synergetic effect of simultaneous exposure to the virus to enhance infectively, severity and fatalities of the disease.

The behavior of an enveloped virus is affected by extra-human conditions, such as climatic conditions (cohen 2020). In this study, solar radiation plays an important role in the propagation of covid-19 in Brazil.

In Iran Almadi et al (2020) found that radiation was inversely related to covid-19 infection, using a linear correlation tool. However, it was not significant. In the present studies, the relationship between these two variables was inverse and significative. A non-linear correlation tool was used to analyzed the data in Rio de Janeiro.

Among weather features, the influence of temperature has been the most studied (Tosepu et al 2020). In this study, temperature (average and maximum) was associated ($p < 0.01$) with lower number of COVID-19 infection. High temperatures damage the virus lipid layer decreasing its stability and infection potential and may even cause virus inactivation, therefore lowering the

transmission rate. In a survival test, (prusin II et al.2018), using an enveloped virus, suggested as “equivalent to influenza and coronavirus”, verified the temperature effect on the virus inactivation. The increase from 14 to 34 °C caused the 6-log reduction of the microorganism studied (prusin II et al). The previous findings corroborate with the significant and negative correlation found for temperature (average and maximum) and the number of cases in Rio de Janeiro.

The wind was implied as critical factor for the transmission of covid-19, however, studies on this factor are scarce. She et al 2020 found that wind speed was a significate factor for the SARS-Cov-2 outbreak in Beijing China. In Rio de Janeiro, higher windspeed were associated with lower incidences of covid-19.

In an external environment, high windspeed contribute to the dilution and removal of the droplets that decreases the concentration of the virus in the air, this reducing the transmission potential of SARS according to WHO 2003 cai et al 2007.

CHAPTER THREE

3.0 Data and Methodology

3.1 Data type and Source

The wind speed and temperature data from Kenya Meteorological Department (KMD) data archive for Eldoret Airport, Moi international airport Mombasa, JKIA and Dagoretti synoptic stations. The data were averaged monthly from March 2020 up to February 2021. They were then averaged seasonally from the season of MAM, JJA, SON, DJF.

The covid-19 data were obtained from the ministry of health official website from which daily covid-19 briefings were available. They were averaged monthly from March 2020 up to February 2021 then later grouped in an average of three month according to the respective seasons in Kenya.

They included:

	DATA TYPE	**TEMPORAL SPAN**	**DATA SOURCE**	**RESOLUTION**
NAIROBI	Covid-19 data	March 2020-Feb 2021	MOH website www.health.go.ke	Averaged monthly and seasonally by the number of reported days
	Weather data	March 2020-Feb 2021	KMD	Averaged seasonally and monthly
MOMBASA	Covid-19 data	March2020-Feb 2021	MOH website www.health.go.ke	Averaged monthly and seasonally by the number of reported days
	Weather data	March 2020-Feb 2021	KMD	Averaged seasonally and monthly
UASIN-GISHU	Covid-19 data	March 2020-Feb 2021	MOH website www.health.go.ke	Averaged monthly and seasonally by

				the number of reported days
	Weather data	March 2020-Feb 2021	KMD	Averaged seasonally and monthly

3.2 Data quality control

Data quality control is the process of controlling the usage of data for an application or process. It ensures that the data meets required standards and involve looking for errors involved, data consistency and uniformity.

The wind speed and temperature data quality is guaranteed due to the implementation of quality management in all meteorological weather providers by ICAO to examine competence and consistency before transmission and storage.

3.2.1 Estimation of missing data

Missing data can be estimated by various methods such as correlation method, arithmetic mean method, isopleth method, isohyetal linear interpolation, arithmetic mean method and thiessen polygon method.

In this case the missing data was found to be less than 10% of the total data, arithmetic mean method was hence used as suitable method of estimating missing less than 10%. The equation below is for arithmetic mean method.

$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{N}$ …… Equation 1

$\bar{x}\ total\ average\ of\ the\ population.$

$x_i\ sum\ of\ the\ population$

Nnumber of the population

3.2.2 Homogeneity test

A Chi-square test was used to check for consistency of temperature and wind speed data. The chi-square is used to determine whether the variables independent or related.

3.3 Methodology

Methods used in the study included scatter plots to observe relationship between wind speed and covid-19 cases in each county in question.

Bar graphs was used to show relationship between weather parameters and covid-19 cases to compare with different regions.

3.3.1 Scatter plots

Scatter plots are useful since they are used to determine between two variables, this relationship can be between two causes or cause effect. Pattern of relationship here is analyzed by changing independent variable and monitoring how it changes the dependent variable.

Covid-19 cases was plotted against temperature and windspeed data for the study area of Nairobi, Mombasa and Uasin-Gishu.

The linear relationship showed the correlation coefficient between covid-19 cases and temperature and wind speed in each study area.

3.3.2 Correlation Analysis

Correlation analysis was used to determine the degree by which two variables are related. It also used correlation coefficient through which we know how much one variable change when the other one changes.

The relationship between the covid-19 cases and temperature, and the

relationship between covid-19 cases and wind speed in the three study areas, was calculated to show how variables changes with respect to each other. The Pearson correlation coefficient R was calculated to show the strength of their relationship.

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

equation 2

Where

Xi, Yi are individual sample points, are sample means

r is the range -1 < r > 1.

Positive r shows positive relationship

Negative shows negative relationship

0 shows no relationship.

Significant testing for computed correlation coefficient was done using the t test at 95% level of significance according to the below formula:

$t = r, \sqrt{\frac{n-2}{1-r^2}}$ equation3

Where n-1 is the degree of freedom, r is the correlation between x and y is the coefficient of determination.

The hypothesis of testing significance was;

Null hypothesis; Ho: correlation between weather parameters and covid-19 prevalence in Kenya is zero.

Alternative hypothesis; H1: correlation between weather parameters and covid-19 prevalence in Kenya is not equal to zero.

Tabulated T at a significant level of 0.05 and degrees of freedom of (12-1) and (4-1) was also obtained.

If T computed is greater than T tabulated, the null hypothesis is rejected and alternative hypothesis accepted.

3.3.3 Regression

Is used to determine which variable impact on topic of interest. Also help to determine which factor matters most, which factor can be ignored and which factor influence each other.

Here regression analysis was used to determine how temperature and windspeed had impacted on the number of covid-19 cases in Nairobi, Mombasa and Uasin-Gishu counties. It helped determine to what extent windspeed and temperature affected the spread of covid-19 cases in the three study counties.

This linear regression equation was used:

$Yi = f(Xi, ß) + ei$ .....Equation 4

Where

Yi =dependent variable

f =function

Xi =independent variable

ei =error terms

ß =unknown parameter

the equation has the form

Y = a + bx

Where:

Y is the dependent variable

a is the y-intercept

b is the slope

x is the independent variable

CHAPTER FOUR

4.0 RESULTS AND DISCUSSIONS

The various results obtained from the data analysis are illustrated in this chapter

Present results in the order of specific objectives:

i. Determine the spatio-temporal distribution of weather

ii. Determine the spatio-temporal distribution of covid prevalence

iii. Establish the relationship between weather and covid prevalence

4.1 Result for correlation between monthly covid-19 prevalence and windspeed

Relationship between covid-19 prevalence per month and monthly wind speed Uasin gishu county was subjected to correlation analysis and a scatter plot was drawn as shown in the figure below

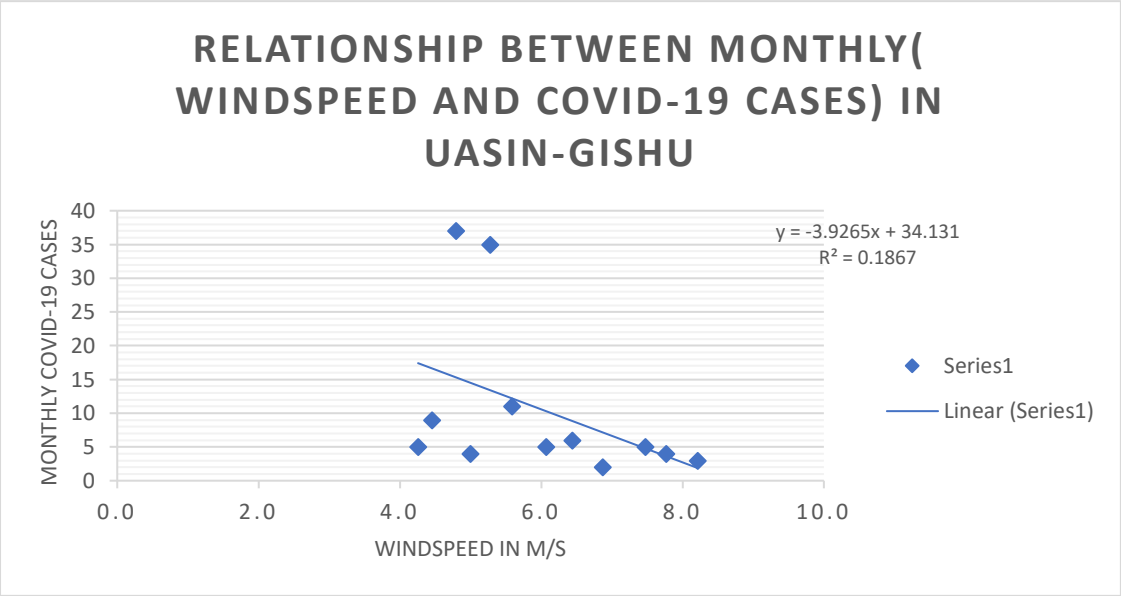

Figure 6: Relationship between monthly windspeed and corona prevalence in uasingishu

From the graph it was found that there was a negative correlation between wind speed and covid-19 cases monthly in that higher wind speed recorded low cases while lower wind speeds recorded higher cases in this county. The graph shows a relationship. The relationship has correlation coefficient of 0.4321, which might show a weak relationship due lack of transparency in reporting covid-19 cases by the county but it shows a moderate association which can be a hint that wind speed in Uasin-Gishu county contributed to its low number of covid-19 cases. The relationship between wind speed and covid-19 in Uasin-Gishu was subjected to a significance testing, T computed was found to be 1.558 and T computed was 1.793885 at 0.05 degrees of freedom and 11 level of confidence. T tabulated is greater than T computed hence we accept the null hypothesis. However according to the past studies, the risk of accepting the null hypothesis may be as a result of type 2 error.

4.2 Result from temporal variation between covid-19 and windspeed in Uasin-Gishu

The figure below shows how covid-19 prevalence and wind speed varied with time for the four seasons

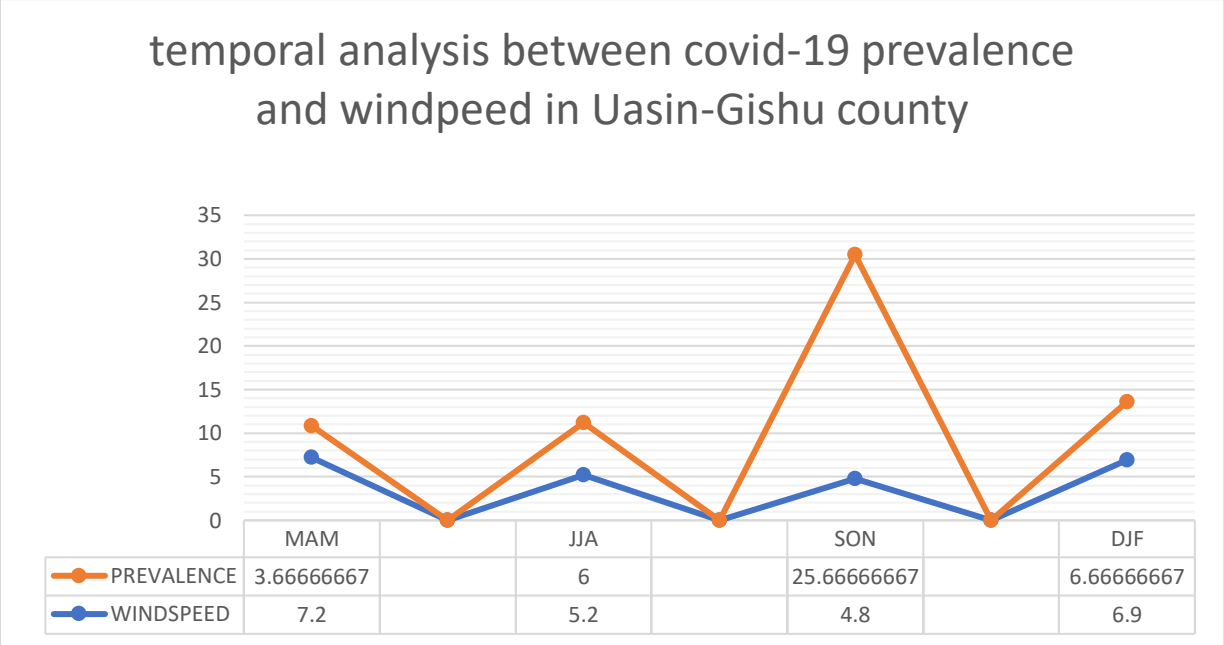

Figure 7: graph for temporal variation between corona prevalence and wind speed in Uasin-Gishu

The figure above shows that the highest wind speed of 7.2m/s in average was recorded in the first season and it recorded the least number of corona case which was only an average of 3.666 in that season. The least wind speed was seen in the third season which was 4.8m/s on average for that season and it recorded the highest corona cases which was the highest in the year, it was averaged to 26 cases for that season. This showed that higher wind speed have the potential to reduce the number of covid-19 case while lower windspeed have the capacity to increase the number of covid-19 case.

Significantly, during MAM, there was highest wind speed of 7.2m/s it recorded the lowest cases of 3.6667, the wind speed dropped to 5.2m/s in JJA and the cases increased to 6, when the windspeed further dropped to 4.8m/s, the cases increased up to 25.667 but when the wind speed increased again up to 6.9 the cases dropped from 25.667 to 6.6667.

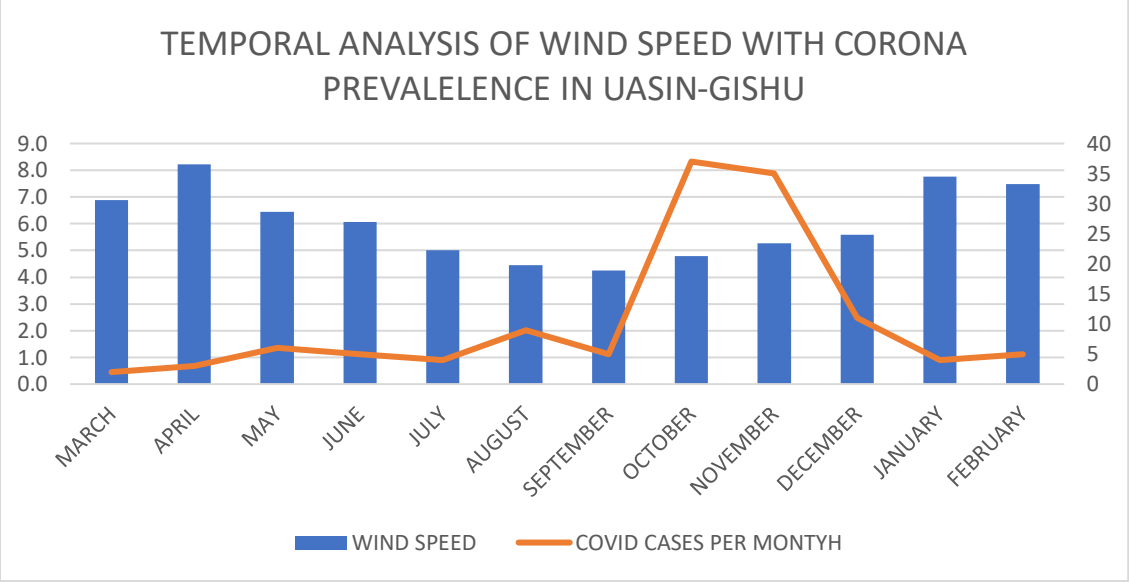

Figure 8: combine line and bar graph showing temporal analysis for wind speed and corona prevalence in Uasin-Gishu

The graph above shows that from march to July the there was low prevalence as wind speed was trailing high. The wind speed started decreasing from July to December and the prevalence also started a surge up to December. From December the surge was reducing as the wind rises in speed up to February.

4.3 Result from correlation between corona wind speed seasonally

scatter plot was used to show this relationship as show in the figure below

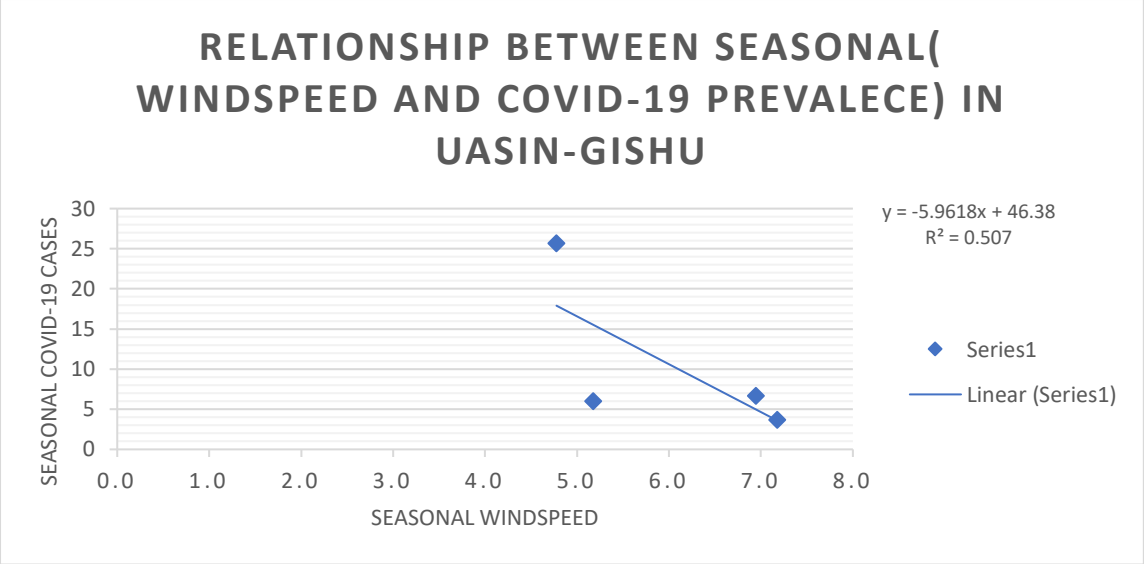

Figure 9: relationship between seasonal windspeed and corona prevalence in Uasin-Gishu**.**

From the graph of relationship between seasonal windspeed and covid-19 prevalence in Uasin-Gishu county, the relationship was found to have a correlation coefficient of 0.71204 which suggest a strong relationship between covid-19 cases and windspeed in each season annually. This that wind speed contributed immensely to reported corona cases in the county in question when it was given a seasonal approach, and that the effect of windspeed on corona cases in Uasin-Gishu could only be realized after every four months period. When test for significance was conducted, t computed was 1.4356 and t tabulated was 2.353363 at 0.05 level of confidence and 2 degrees of freedom. t tabulated was greater than t computed we therefore accept null hypothesis however the risk of accepting null hypothesis may be as a result of type 2 error.

4.4 Result from correlation analysis of monthly average temperature and covid-19 cases in Uasin-Gishu

After successful data collection and analysis, the following figure show the relationship between monthly temperatures and covid-19 cases

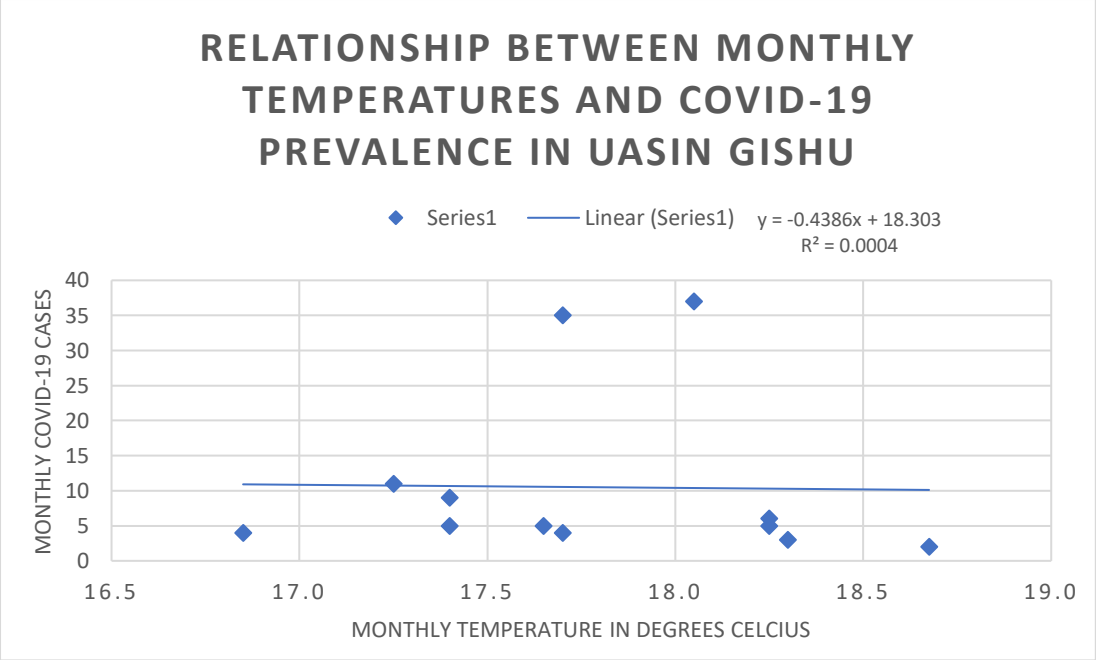

Figure 10: graph for the relationship between monthly temperature and corona prevalence in Uasin-Gishu.

From the graph it shows that the correlation coefficient was 0.02 which shows very weak or negligible correlation between temperature and corona cases even though there was a trend, this suggest that there might be no significant contribution of temperature to the reported covid-19 cases in this county therefore the null hypothesis may be rejected for the case of Uasin-Gishu county. Significance testing was conducted and t computed was 0.06326 and t tabulated was 1.793885 at 0.05 level of confidence and 11 degrees of freedom. t tabulated was greater than t computed we therefore accept null hypothesis which states that temperature is not related to covid-19 prevalence in Uasin-Gishu.

4.5 Results from seasonal correlation analysis between covid-19 prevalence and temperature in Uasin-Gishu county

when data for monthly covid-19 prevalence and monthly temperature in Uasin-Gishu county was averaged into seasonal aspects so as to give it seasonal outlook the results became the following.

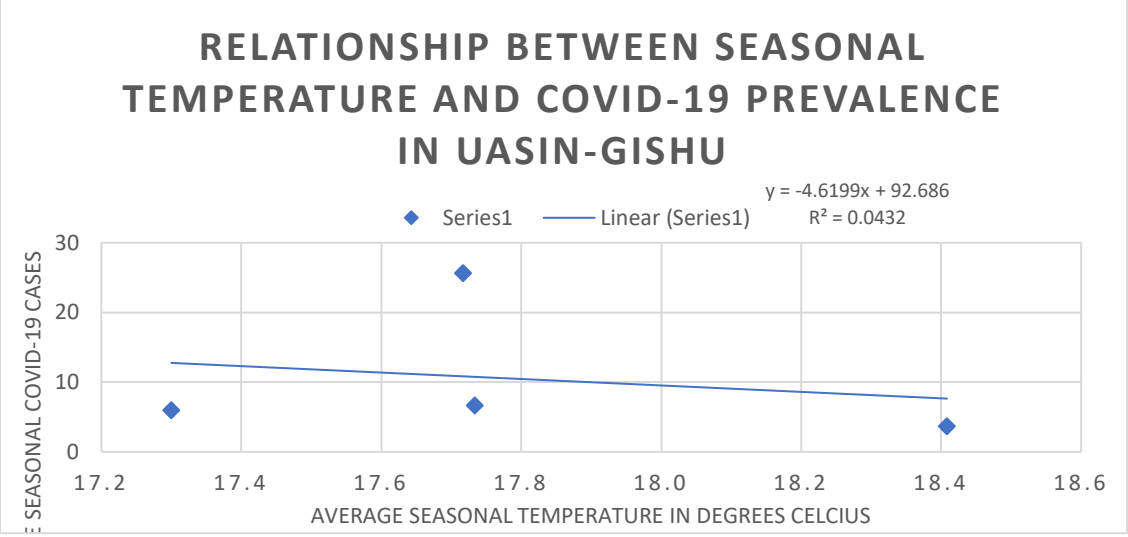

Figure 11: relationship between seasonal temperature and covid-19 prevalence in Uasin-Gishu.

The correlation coefficient became only 0.2078 suggesting a week relationship between temperature and covid-19 prevalence in the county, however there was a trend.

4.6 Correlation analysis between monthly covid-19 prevalence in Mombasa and wind speed.

Scatter plot was drawn and correlation analysis was conducted to check relationship between monthly covid-19 cases and wind speed in Mombasa and the following results were obtained

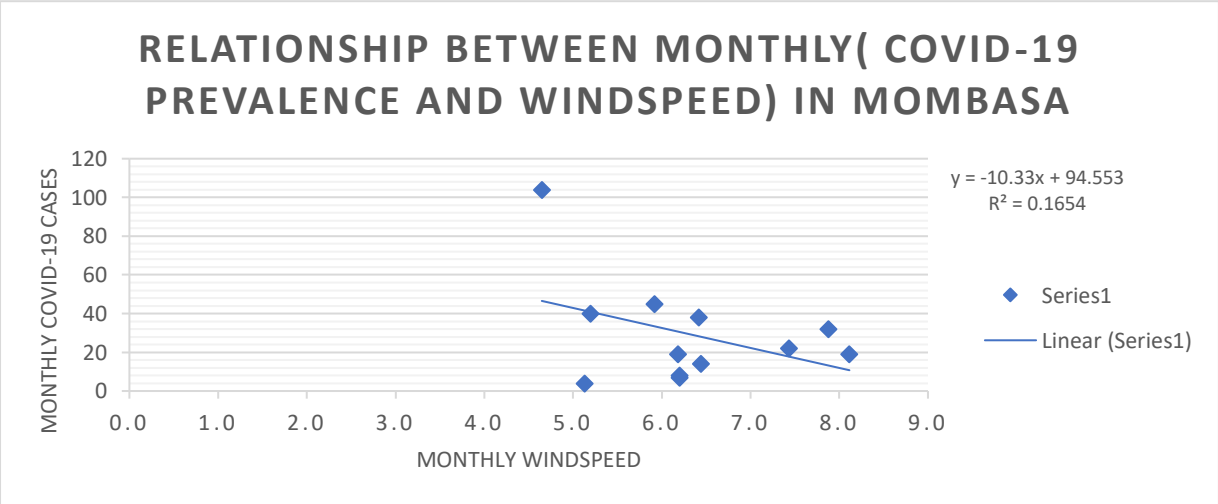

Figure 12: relationship between monthly corona and windspeed in Mombasa.

The correlation coefficient was found to be 0.4067. this relationship although considered to be weak, it has a moderate association and a positive trend. This suggest that windspeed aspect might have brought somehow little impact on the cases that were being reported monthly in Mombasa county. Test for significance was done where t computed was 1.4067 and t tabulated

was 1.812461 at 0.05 level of confidence and 11 degrees of freedom. t tabulated was greater than t computed hence we accept the null hypothesis. However correlation coefficient of 0.4067 show some moderate association.

4.7 Relationship between seasonal averaged covid-19 prevalence and windspeed in Mombasa.

The following are the results for the correlation analysis when covid-19 cases and wind speed in Mombasa county were averaged seasonally and a relationship drawn.

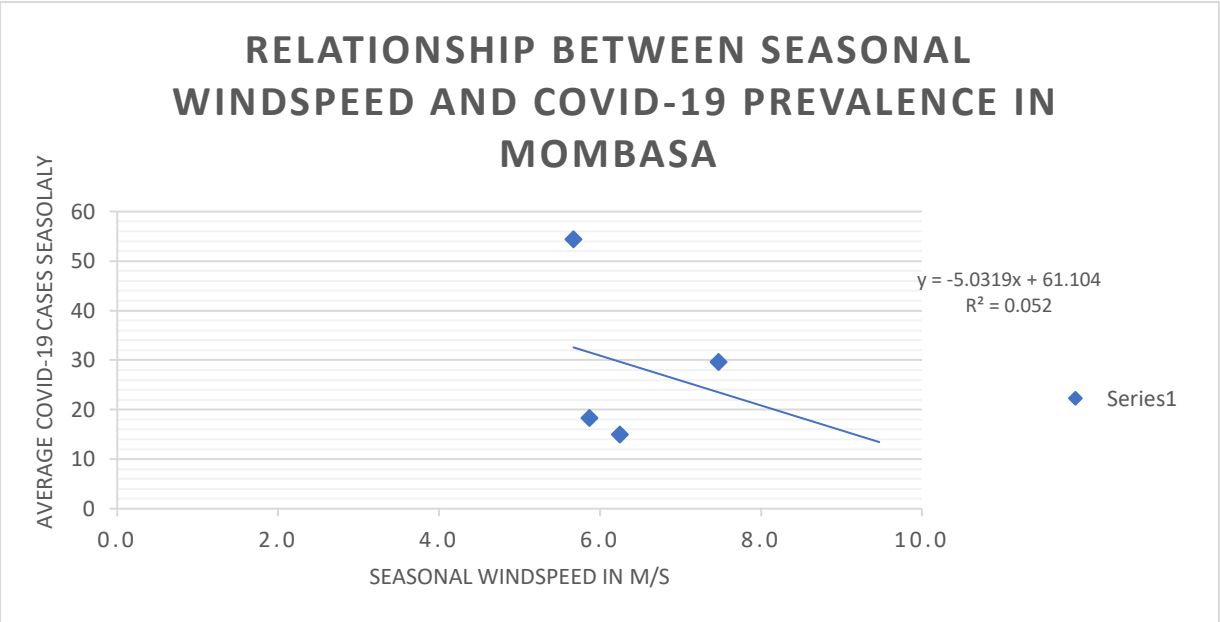

Figure 13: relationship between seasonal windspeed and covid-19 prevalence in Mombasa**.**

The correlation was found to 0.228 which is a very weak relationship between cases and the wind speed. This means that in Mombasa county despite it high average wind speed 6.3 m/s ,this windspeed contributed very little or had a little significance on covid-19 cases reported.

4.8 Results from correlation analysis between covid-19 prevalence and temperature in Mombasa

After analysis the following results were drawn from relationship between seasonally averaged covid-19 cases and temperature.

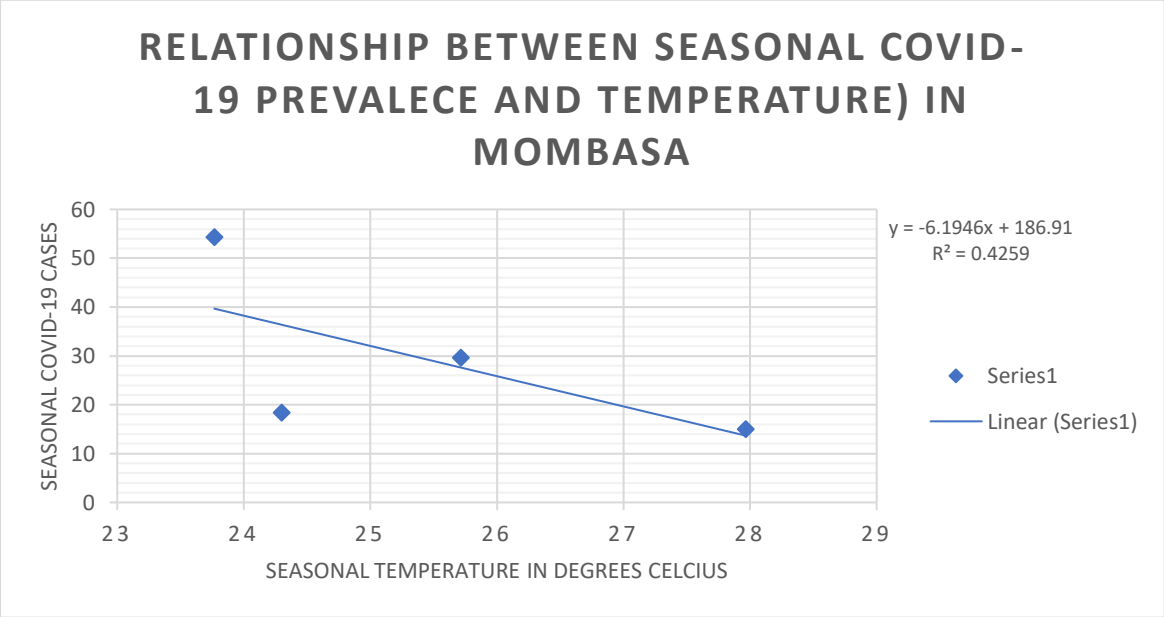

Figure 14: relationship between seasonal covid-19 and temperature in Mombasa.

The correlation coefficient in this case was found to be 0.6526 which shows a very strong relationship between seasonally averaged covid-19 prevalence in Mombasa and temperature. This shows that the rate of infection in the county of Mombasa might have been accelerated by temperature variability. Test for significance was conducted, t computed was found to be 1.2181 and t tabulated was found to be 2.9199 at 0.05 level of confidence and 2 degrees of freedom. t tabulated was greater than t computed we therefore accept null hypothesis. However, according to past studies, the risk of accepting the null hypothesis maybe as a result of type 2 error.

4.9 Temporal analysis between averaged seasonal covid-19 prevalence and temperature in Mombasa county

The cases of corona versus temperature in Mombasa county were looked into overtime so as to see their temporal relationship. The figure below shows that relationship.

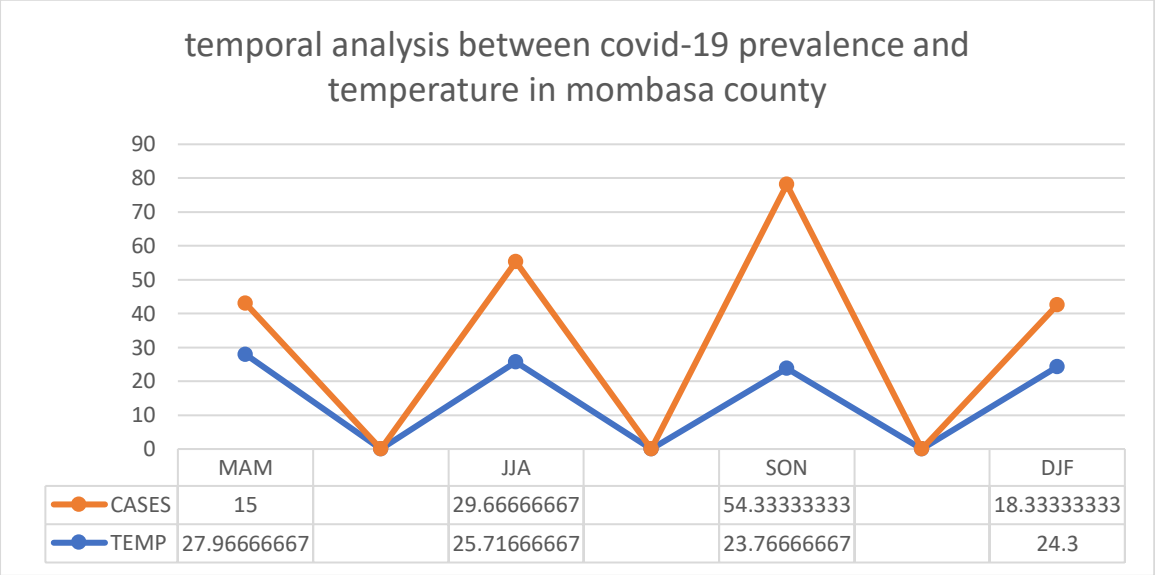

Figure 15: temporal analysis between covid-19 prevalence and temperature in Mombasa**.**

From the figure 13 above it clearly shows that temperature had some impact on covid-19 cases in the coastal city of Mombasa.

From the figure it shows that the lowest temperature which was recorded in SON, 23.7 degrees centigrade recorded the highest corona cases during that period which was, 54.33. on the other hand, the highest temperature on average for the season of MAM which was 28.0 degrees centigrade had only 15 corona cases.

Also, when there is temperature increase from 23.7 to 24.3 there is a corresponding decrease in covid-19 cases from 54 to 18, although it is not a regular decrease but it might have responded to increase in the temperature since it was earlier found that there is a strong relationship between the two variables.

Significantly it can be seen that during MAM temperature of 27.9667 had 15 cases, when the temperature dropped to 25.71667 the cases increased to 29.6667 in JJA, the temperature further dropped in SON up to 23.76667 and the cases again increased up to 54.3333, in DJF, the temperature increased up to 24.3 and the cases reduced up to 18.1333.

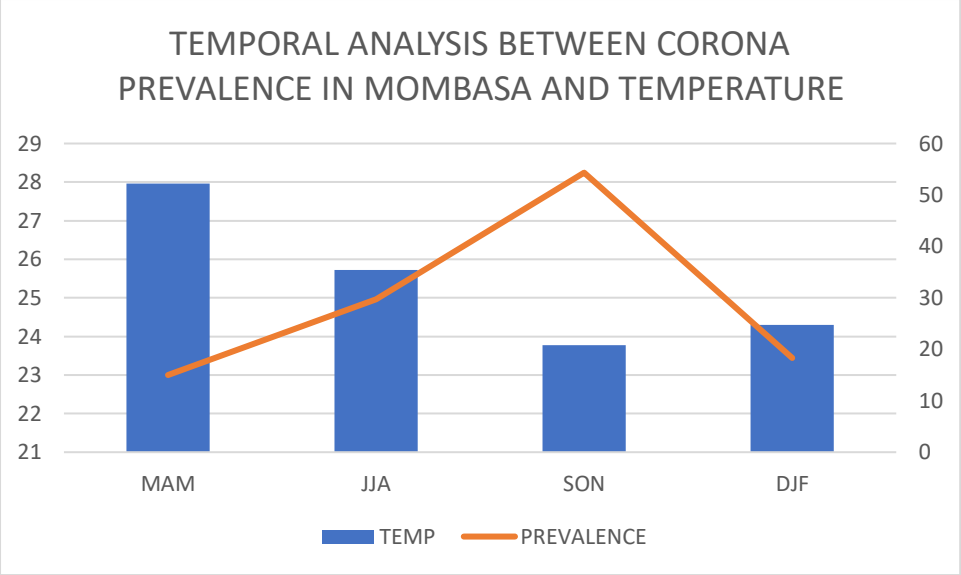

Figure 16: temporal analysis between corona prevalence in Mombasa and temperature.

From the graph it can be deduced that the season of MAM had the highest temperature in Mombasa and it reported low prevalence. JJA had a reduced temperature and the prevalence started rising and reached peak in the season of SON which also had the lowest temperature.

4.10 Results for the correlation analysis of monthly covid-19 prevalence and wind speed in Nairobi county.

In the figure below are the results for the relationship between average monthly covid-19 cases and wind speed in the county of Nairobi.

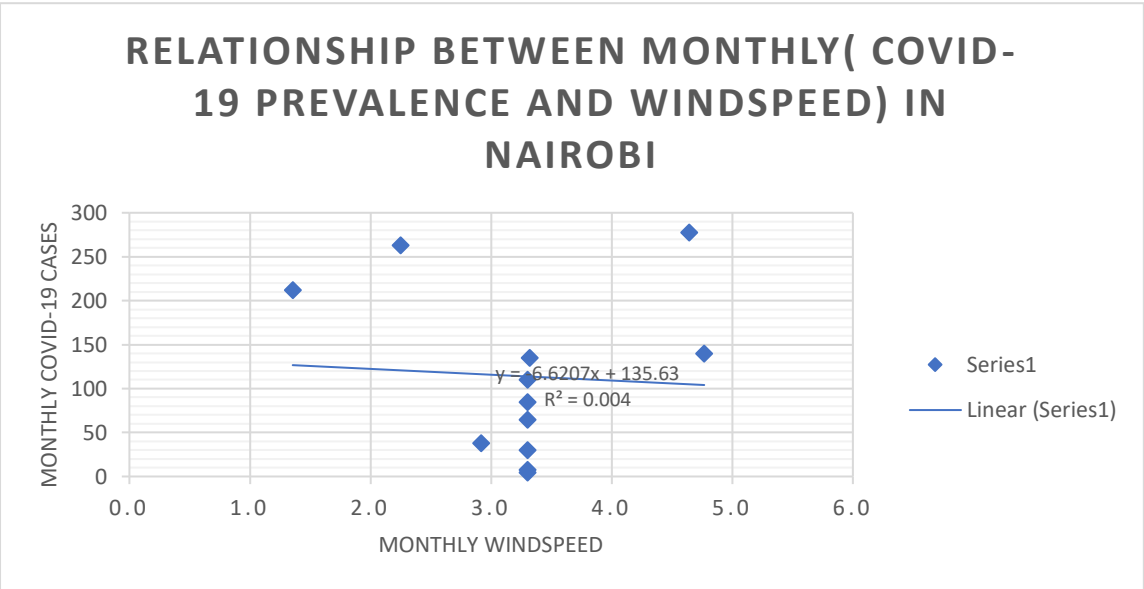

Figure 17: relationship between monthly covid-19 prevalence and windspeed in Nairobi.

From the figure 16 above shows a trend in the relationship between monthly covid-19 cases and windspeed in Nairobi county, however the relationship is weak since it has a correlation coefficient of 0.0632. this shows that the wind speed in the county of Nairobi might have contributed very little or none to the monthly cases that were reported in the city. Significance test was conducted, t computed was 0.2003 and t tabulated was 1.812461 at 0.05 level of confidence and 11 degrees of freedom. t tabulated is greater than t computed and we therefore accept the null hypothesis which state that correlation between covid and wind in Nairobi is zero.

4.10 Results for correlation analysis between monthly temperature and covid-19 in Nairobi county.

The figure below displays the correlation analysis that was obtained in monthly covid-19 cases and temperature relationship.

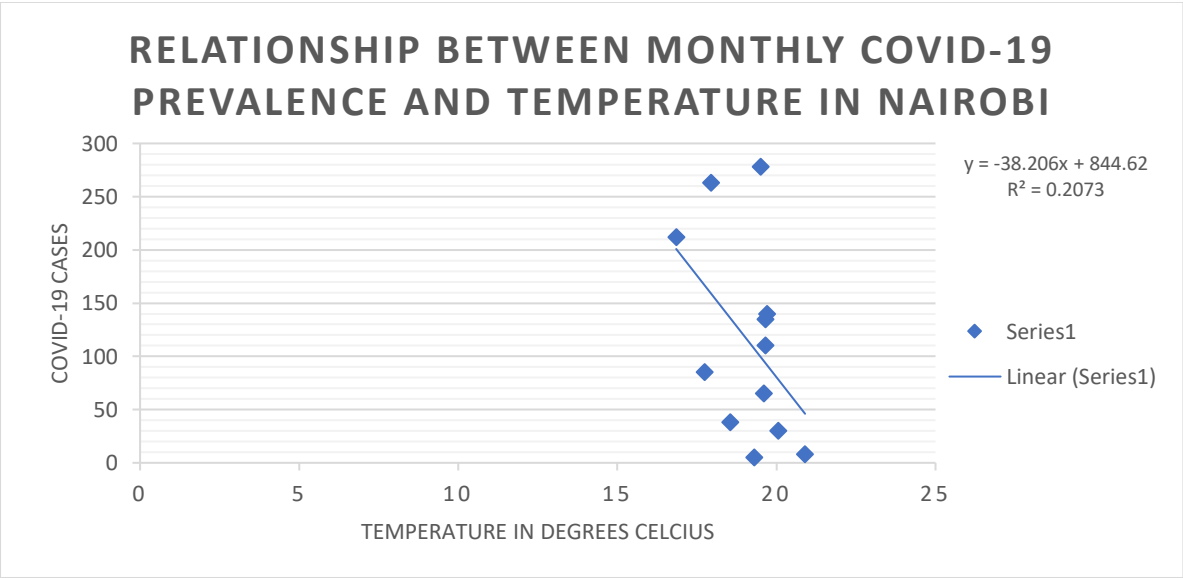

Figure 18: relationship between monthly covid-19 prevalence and temperature in Nairobi.

The figure below shows a correlation coefficient of 0.4553 which is regarded as a weak relationship. Although weak, it is moderately weak and also shows a moderate association. This suggest that temperature might have contributed to monthly reported covid-19 cases to some extent in the city of Nairobi. The significance test was conducted where t computed was found to be 1.617 and t tabulated was 1.812461 at 0.05 level of confidence and 11 degrees of freedom. t tabulated was greater than t computed and we therefore accept the null hypothesis however the risk of accepting the null hypothesis may be as a result of type two error.

4.11 Result for the correlation analysis between averaged seasonal temperature and covid-19 prevalence in the city of Nairobi.

The figure below shows the results for the correlation analysis between seasonal temperature versus covid-19 cases displayed in a scatter plot.

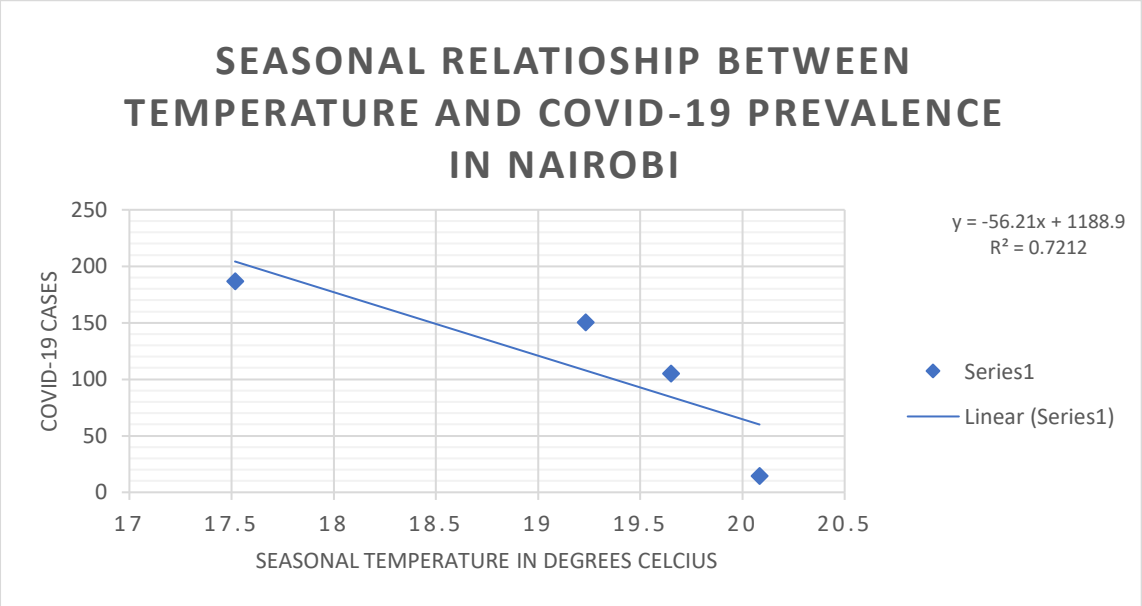

Figure 19: seasonal relationship between temperature and covid-19 prevalence in Nairobi.

In the graph above it clearly show a correlation coefficient of 0.8492, this shows that temperature was strongly related to the number of covid-19 cases in Nairobi. The effect of temperature on covid-19 prevalence in Nairobi was realized after every four months period, every season the effect would display. The test for significance was conducted and t computed was found to be 2.9212 and t tabulated was 2.9199 at 0.05 level of confidence and 2 degrees of freedom. t computed was greater than t tabulated. We therefore reject the null hypothesis which state that relationship between weather parameter and corona prevalence is zero and accept the alternative hypothesis.

4.12 Results for the temporal analysis between seasonal covid-19 cases and temperature in Nairobi county.

The temporal analysis was conducted to investigate the trend for the relationship between covid-19 and temperature seasonally for the city of Nairobi and the following results were obtained.

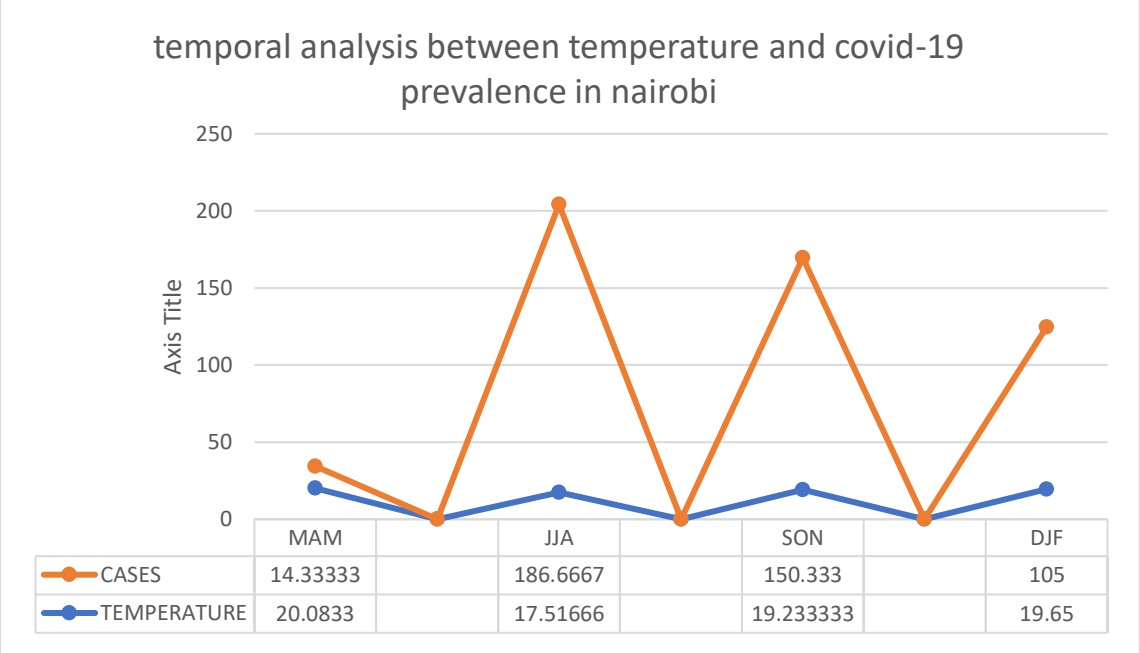

Figure 20: temporal analysis between temperature and covid-19 prevalence in Nairobi.

From the graph it shows that at temperature 20.08, the cases were 14.33, at temperature 17.52, the cases were 186.67, at temperature 19.23, the cases were 150.33. we can see that the lowest amount of temperature which was 17 degrees centigrade recorded the highest cases which was averaged to 186.667 cases while the highest amount recorded the least cases.

Significantly, it can be seen that the highest temperature of 20.0833 degrees during MAM recorded the lowest temperature of 14.333, when it dropped to 17.5166 the cases increases to 186.6667 in JJA, the temperature increased to 19.2333 in SON the cases again dropped from 186 to 150.3333. in DJF where the temperature slightly increased by 0.4 the there was a slight drop in case by 45.

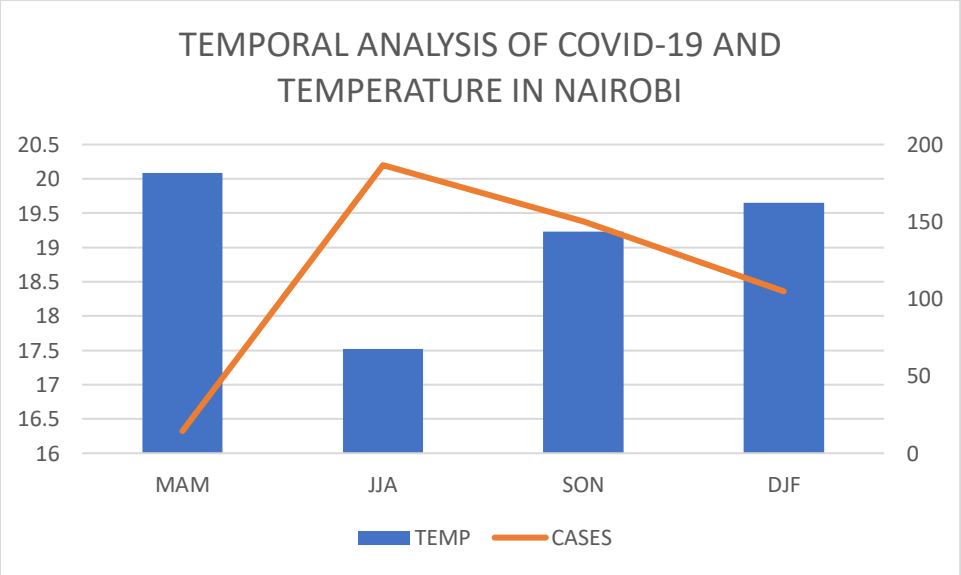

Figure 21: temporal analysis of covid-19 and temperature in Nairobi.

From the figure 20 above it can be seen that MAM in Nairobi was the season of the highest temperature received the lowest prevalence. The prevalence in JJA was the peak and the same season had the lowest temperature. As the temperature started increasing from SON all the way to DJF the prevalence was also seen to lower.

4.13 Results from the regression analysis between Covid-19 cases and weather parameters.

Fitting a simple linear regression model, regression models $y = -56.21x + 1188.9$ for the correlation between temperature and covid-19 cases in Nairobi, and $y = -6.1946x + 186.91$ for the correlation between temperature and covid-19 obtained from Mombasa, and $y = -5.9618x + 46.31$ for the correlation between windspeed and covid-19 cases obtained from Uasin-Gishu. These three models have significant R^2 of 0.7212, 0.4259, and 0.509 respectively. From the regression analysis. We clearly see significant contribution of weather parameters to covid-19 cases. When all factors are held constant, temperature is inversely proportional to the number of covid-19 and so windspeed.

CHAPTER FIVE

5.0 Summary, Conclusion and Recommendation

This chapter provide a rap up of the study, deductions drawn from the results of the study and recommendations for the various findings of the study.

5.1 Summary

The main objective of the study was to investigate the effect of weather parameters on the covid-19 cases reported in the country of Kenya where three counties of Nairobi, Mombasa and Uasin-Gishu were considered for the study. The county of Nairobi and Mombasa were picked to investigate the effects of weather parameters on corona cases in the urban area while the county of Uasin-Gishu was to investigate the effect on rural area.

For the case of Uasin-Gishu, it was found that the wind speed had a significant influence on corona cases which could be more realized after every season. The strong relationship of 0.71204 between those two variables suggested that wind must have fueled the spread of covid-19 in this county. On the other hand, temperature had no significant relationship with the cases of covid-19 since the correlation coefficient between the two variables were only 0.2076 on a seasonal outlook.

In the county of Mombasa which was secondly investigated, it was found from the results that wind speed had a weak relationship of 0.228 and 0.4067 with corona virus cases both monthly and seasonally respectively suggesting that wind speed might have not contributed to corona virus cases in Mombasa. However, temperature showed a moderate association with corona cases of which this became a strong relationship of 0.6526 correlation coefficient when it was given a seasonal look. This suggest that the effect of temperature on covid-19 cases would be realized after every season, four months period.

In the county of Nairobi, the effect of wind was not significant to the cases of corona virus following the weak relationship that displayed a correlation coefficient of 0.0632 between them which was displayed by small correlation coefficient that was tending to zero.

The temperature showed a correlation coefficient next to one which was a very strong relationship. The strength of this relationship was more pronounced when seasonal outlook was

considered of which it was 0.8492. This means that temperature must have contributed to the corona virus cases that were reported in Nairobi county.

5.2 Conclusions

The study investigated the influence of weather parameters on corona virus cases in Kenya. Temporal analysis, regression analysis and correlation analysis were used for the study. From the study it shows that temperature affected corona virus cases more in the urban area than in the rural area as evident by the strong relationship between temperature and covid-19 cases in Nairobi county and Mombasa county.

The study showed that increase in the temperature reduces the corona virus cases while decrease in temperature increases the corona virus cases in the urban areas of Kenya. In the temporal analysis both monthly and seasonally, corona virus cases were seen to surge in the cold seasons than in the warm season, a decrease in temperature of the next season resulted to an increase in corona cases in that season and vice versa. The biasness in reporting coronavirus cases effectively in the counties of Kenya might have hindered effective result in the course of the study

Windspeed seems to have influenced corona virus cases in the rural towns than in the urban cities, this was evident when Uasin-Gishu displayed a strong relationship of corona virus cases and wind speed seasonally than Nairobi and Mombasa, however, temperature might have not contributed much as it was evident by weak correlation coefficient between it and the reported cases in the county. Seasons which had a higher wind speed in the county reported lower corona cases whereas seasons which low average windspeed recorded a bit higher corona cases.

Nairobi and Mombasa are urban areas, urban areas have a relatively low windspeed compared to the rural areas, the wind speed in the urban areas are hindered by many factors such as friction from the urban structures and hinderance by lots of activities in the urban areas this might be the reason why wind speed was a weak influence in Nairobi county.

5.3 Recommendations

After investigating the effects of windspeed and temperature on corona virus cases in Nairobi, Mombasa and Uasin-Gishu, there is need that more weather parameters like humidity, amount

of sunshine and even precipitation be investigated and more counties in Kenya be included for wider understanding.

There is need for government to take much care as fare as enforcing covid-19 protocols are concerned during cold seasons in urban areas rather than waiting for the resurgence to enforce the covid-19 protocols, this might help curb the spread

There is need for people in the urban regions also to be informed about the ease of contracting the virus during the cold seasons so that the people may also take cautionary measures during those periods to help reduce the number of infections

Night hours also are accompanied by cold a temperature, hence this study has justified the dusk to dawn curfew enforced by the government in the mostly affected counties as this will help reduce the spread of corona virus in the county.

Installation of heater fans in enclosed places like offices and churches. These heater fans will create low pressure in the rooms leading to more air entering the rooms causing air motion and driving away stagnant air that might have been infested by the virus. The heater fans may also cause the buoyancy.

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